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Topic 14

Sound

Reading Question 14-1 (1 of 2)

Raising the temperature of air will

1. decrease the speed of sound in the air.
2. increase the speed of sound in the air.
3. not affect the speed of sound in the air.
4. sometimes increase and sometimes decrease the speed of sound in air.

Reading Question 14-1 (2 of 2)

Raising the temperature of air will

1. decrease the speed of sound in the air.
- 2. Answer: increase the speed of sound in the air.**
3. not affect the speed of sound in the air.
4. sometimes increase and sometimes decrease the speed of sound in air.

Reading Question 14-2 (1 of 2)

Sound waves may be best described as being

1. longitudinal waves.
2. transverse waves.
3. a combination of longitudinal and transverse waves.
4. analogous to water waves.

Reading Question 14-2 (2 of 2)

Sound waves may be best described as being

- 1. Answer: longitudinal waves.**
2. transverse waves.
3. a combination of longitudinal and transverse waves.
4. analogous to water waves.

Reading Question 14-3 (1 of 2)

If the wavelength of sound is increased by a factor of 3, what happens to its speed and frequency?

1. Its speed increases by a factor of 3, and so does its frequency.
2. Its frequency increases by a factor of 3, but the speed stays the same.
3. Its frequency decreases by a factor of 3, but its speed stays the same.
4. Its speed decreases by a factor of 3, and so does its frequency.

Reading Question 14-3 (2 of 2)

If the wavelength of sound is increased by a factor of 3, what happens to its speed and frequency?

1. Its speed increases by a factor of 3, and so does its frequency.
2. Its frequency increases by a factor of 3, but the speed stays the same.
- 3. Answer: Its frequency decreases by a factor of 3, but its speed stays the same.**
4. Its speed decreases by a factor of 3, and so does its frequency.

Reading Question 14-4 (1 of 2)

Let's say that you are spinning a fixed-frequency buzzer attached to a string in a circular path around you at a constant speed. The frequency that you hear as you spin the buzzer

1. is greater than the frequency of the buzzer and it is constant.
2. alternates between being greater than the frequency of the buzzer and less than the frequency of the buzzer.
3. is less than the frequency of the buzzer and it is constant.
4. is the same as the frequency of the buzzer.

Reading Question 14-4 (2 of 2)

Let's say that you are spinning a fixed-frequency buzzer attached to a string in a circular path around you at a constant speed. The frequency that you hear as you spin the buzzer

1. is greater than the frequency of the buzzer and it is constant.
2. alternates between being greater than the frequency of the buzzer and less than the frequency of the buzzer.
3. is less than the frequency of the buzzer and it is constant.
4. **Answer: is the same as the frequency of the buzzer.**

Reading Question 14-5 (1 of 2)

A ringing alarm clock is placed in a bell jar, which is then sealed, and the air is pumped out of it. Which effect will most likely occur?

1. The sound of the ringing alarm clock will get louder as the air is pumped out.
2. The sound of the ringing alarm clock will get softer as the air is pumped out.
3. No noticeable change will occur as the air is pumped out.
4. The sound will diminish completely immediately after the air has begun to be pumped out.

Reading Question 14-5 (2 of 2)

A ringing alarm clock is placed in a bell jar, which is then sealed, and the air is pumped out of it. Which effect will most likely occur?

1. The sound of the ringing alarm clock will get louder as the air is pumped out.
- 2. Answer: The sound of the ringing alarm clock will get softer as the air is pumped out.**
3. No noticeable change will occur as the air is pumped out.
4. The sound will diminish completely immediately after the air has begun to be pumped out.

Reading Question 14-6 (1 of 2)

When a standing wave occurs on a string,

1. all points along the string move up and down equal distances.
2. all points along the string move up and down, but some move more than others.
3. many points along the string move up and down, some move more than others, and some are stationary.
4. all points along the string are completely stationary because the waves from each direction cancel each other out.

Reading Question 14-6 (2 of 2)

When a standing wave occurs on a string,

1. all points along the string move up and down equal distances.
2. all points along the string move up and down, but some move more than others.
3. **Answer: many points along the string move up and down, some move more than others, and some are stationary.**
4. all points along the string are completely stationary because the waves from each direction cancel each other out.

Topic 14: Sound

TOPIC DISCUSSION

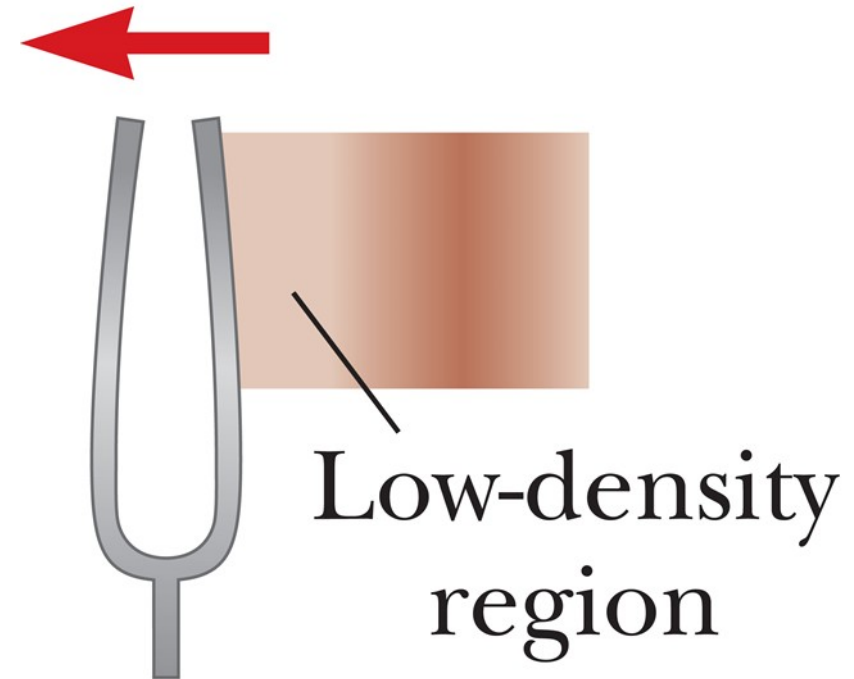
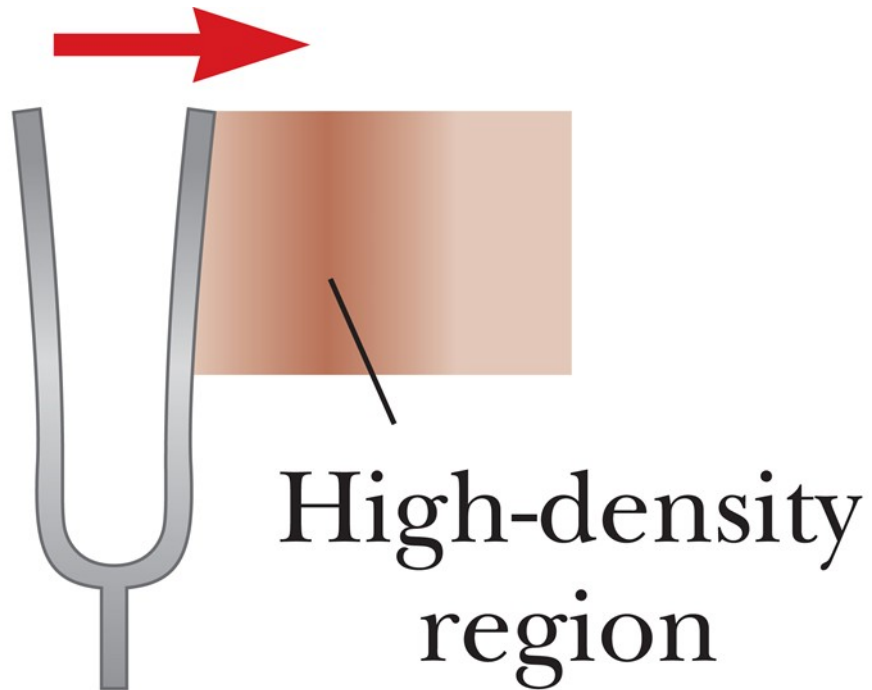
Producing a Sound Wave (1 of 3)



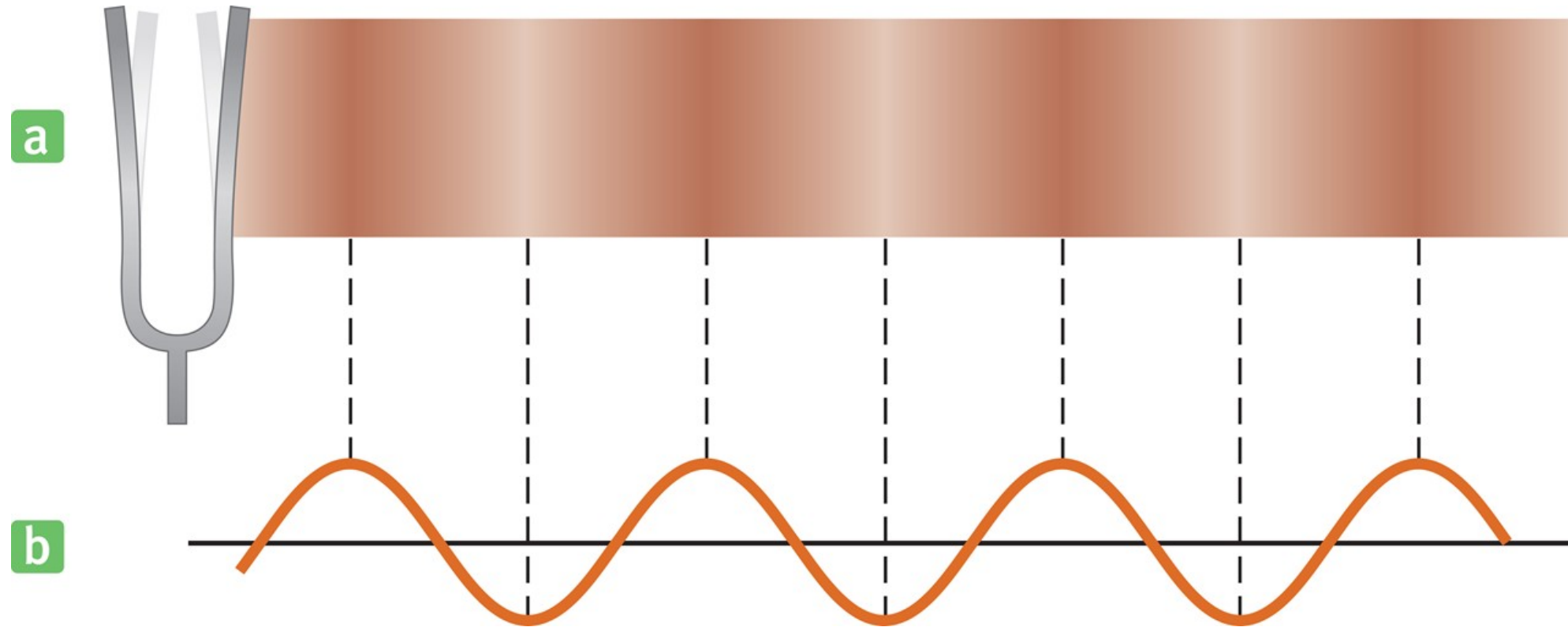
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a

Producing a Sound Wave (2 of 3)



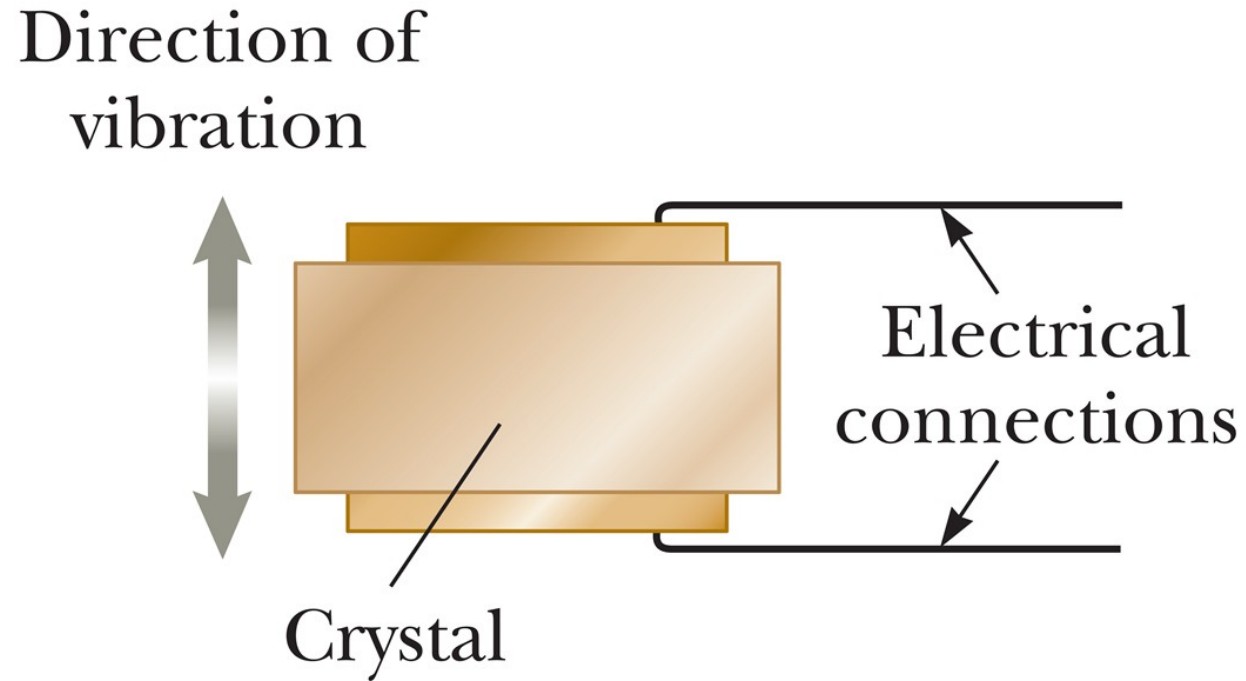
Producing a Sound Wave (3 of 3)



Categories of Sound Waves

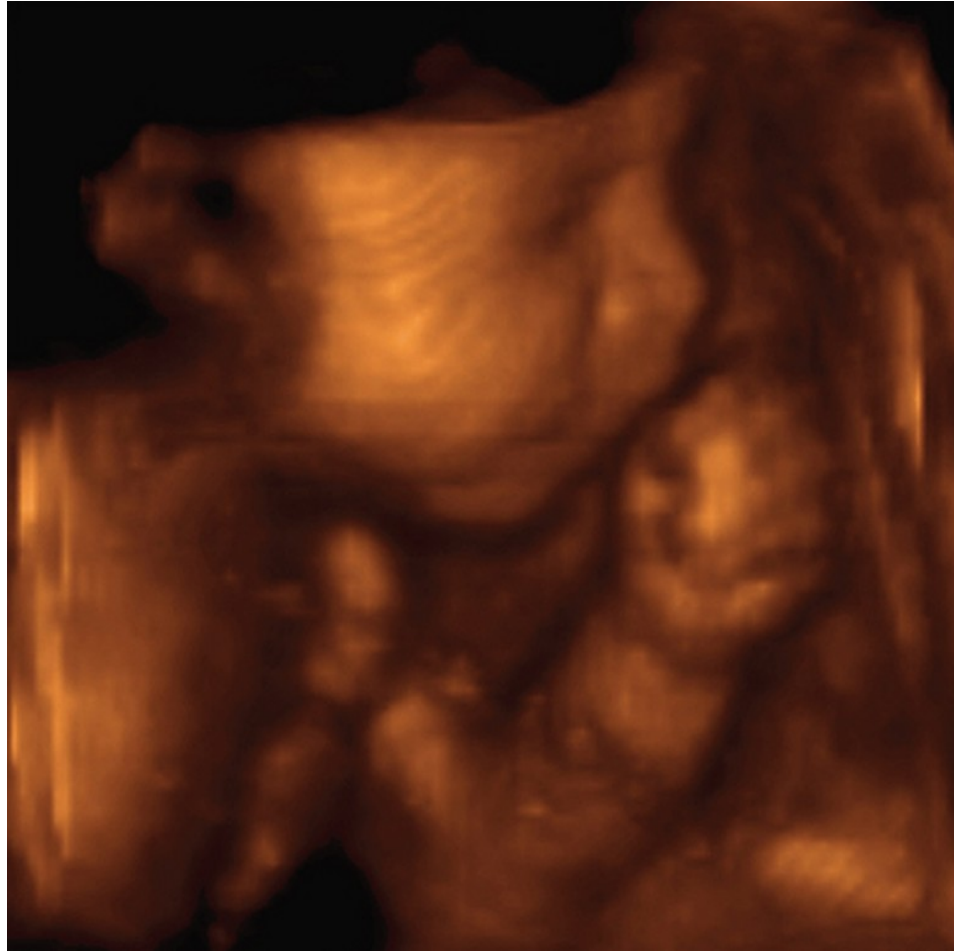
- **Audible waves**
- **Infrasonic waves**
- **Ultrasonic waves**

Applications of Ultrasound (1 of 2)



$$PR = \left(\frac{\rho_i - \rho_t}{\rho_i + \rho_t} \right)^2 \times 100$$

Applications of Ultrasound (2 of 2)



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The Speed of Sound (1 of 4)

$$v = \sqrt{\frac{B}{\rho}} \quad B \equiv -\frac{\Delta P}{\Delta V/V}$$

$$v = \sqrt{\frac{\text{elastic property}}{\text{inertial property}}}$$

$$v = \sqrt{\frac{F}{\mu}} \quad (\text{transverse wave on a string})$$

$$v = \sqrt{\frac{Y}{\rho}} \quad (\text{longitudinal wave in a rod})$$

The Speed of Sound (2 of 4)

Table 14.1 Speeds of Sound in Various Media

Medium	v (m/s)
Gases	
Air (0°C)	331
Air (100°C)	386
Hydrogen (0°C)	1 286
Oxygen (0°C)	317
Helium (0°C)	972
Liquids at 25°C	
Water	1 493
Methyl alcohol	1 143
Sea water	1 533

The Speed of Sound (3 of 4)

Medium	v (m/s)
Solids^a	
Aluminum	6 420
Copper (rolled)	5 010
Steel	5 950
Lead (rolled)	1 960
Synthetic rubber	1 600

^aValues given are for propagation of longitudinal waves in bulk media. Speeds for longitudinal waves in thin rods are smaller, and speeds of transverse waves in bulk are smaller yet.

The Speed of Sound (4 of 4)

$$v = (331 \text{ m/s}) \sqrt{\frac{T}{273 \text{ K}}}$$

for $T = 293 \text{ K}$ (room temperature) :

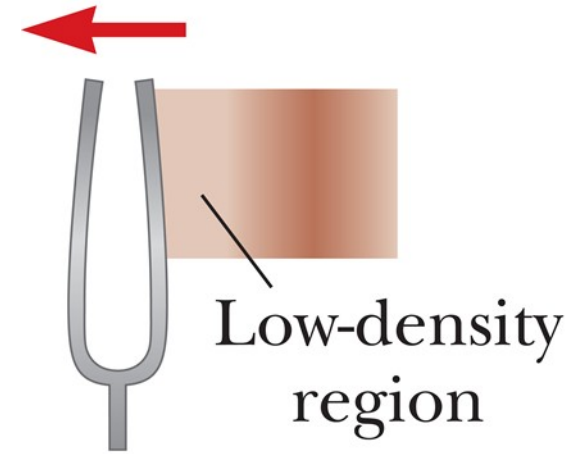
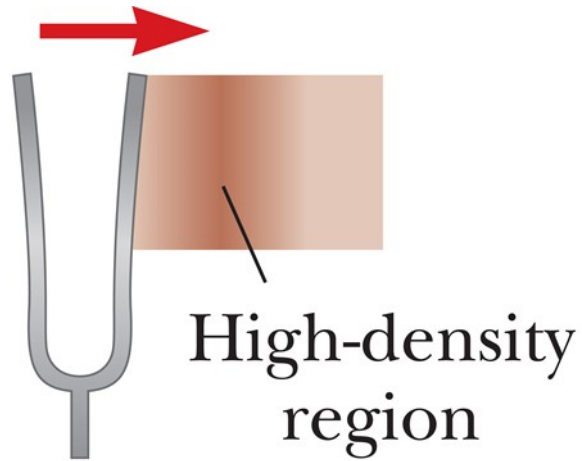
$$v = (331 \text{ m/s}) \sqrt{\frac{293 \text{ K}}{273 \text{ K}}} = 343 \text{ m/s}$$

Think – Pair – Share 1

Which of the following actions will increase the speed of sound in air?

1. decreasing the air temperature
2. increasing the frequency of the sound
3. increasing the air temperature
4. increasing the amplitude of the sound wave
5. reducing the pressure of the air

Energy and Intensity of Sound Waves (1 of 2)



$$I \equiv \frac{1}{A} \frac{\Delta E}{\Delta t} \quad \text{SI unit: W/m}^2$$

$$I \equiv \frac{P}{A} = \frac{\text{power}}{\text{area}}$$

$$I = 1 \times 10^{-12} \text{ W/m}^2 \quad (\text{threshold of hearing})$$

$$I = 1 \text{ W/m}^2 \quad (\text{threshold of pain})$$

Energy and Intensity of Sound Waves (2 of 2)

threshold of hearing →

pressure increase $\approx 3 \times 10^{-5}$ Pa over $P_{\text{atm}} = 1 \times 10^5$ Pa

→ ear detects pressure fluctuations of 3 parts in 10^{10} !

maximum displacement of air molecule $\approx 1 \times 10^{-11}$ m

diameter of molecule $\approx 1 \times 10^{-10}$ m

threshold of pain →

pressure variation ≈ 29 Pa over P_{atm}

maximum displacement of air molecule $\approx 1 \times 10^{-5}$ m

Intensity Level in Decibels (1 of 3)

$$\beta = 10 \log \left(\frac{I}{I_0} \right) \quad I_0 = 1.0 \times 10^{-12} \text{ W/m}^2$$

Intensity Level in Decibels (2 of 3)

threshold of hearing: $\beta = 10 \log \left(\frac{1.0 \times 10^{-12} \text{ W/m}^2}{1.0 \times 10^{-12} \text{ W/m}^2} \right) = 10 \log(1) = 0 \text{ dB}$

$$\beta = 10 \log \left(\frac{1.0 \times 10^{-11} \text{ W/m}^2}{1.0 \times 10^{-12} \text{ W/m}^2} \right) = 10 \log(10) = 10 \text{ dB}$$

$$\beta = 10 \log \left(\frac{1.0 \times 10^{-10} \text{ W/m}^2}{1.0 \times 10^{-12} \text{ W/m}^2} \right) = 10 \log(100) = 20 \text{ dB}$$

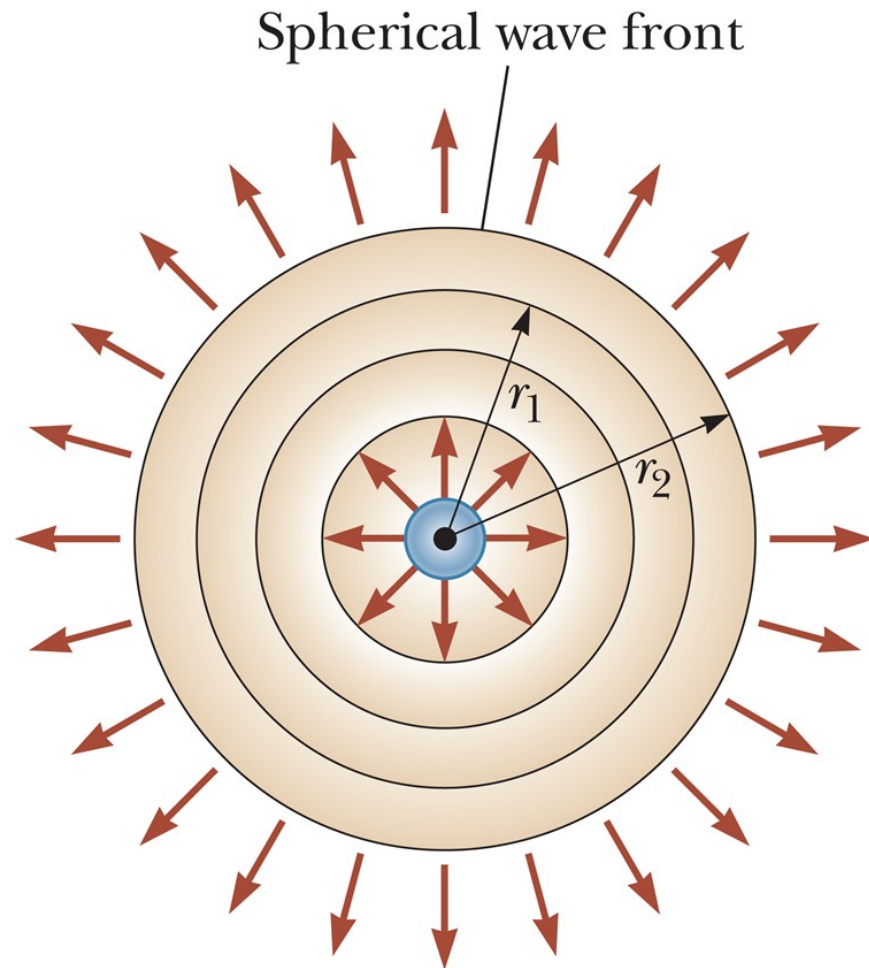
threshold of pain: $\beta = 10 \log \left(\frac{1}{1.0 \times 10^{-12}} \right) = 10 \log(10^{12}) = 120 \text{ dB}$

Intensity Level in Decibels (3 of 3)

Table 14.2 Intensity Levels in Decibels for Different Sources

Source of Sound	β (dB)
Nearby jet airplane	150
Jackhammer, machine gun	130
Siren, rock concert	120
Subway, power mower	100
Busy traffic	80
Vacuum cleaner	70
Normal conversation	50
Mosquito buzzing	40
Whisper	30
Rustling leaves	10
Threshold of hearing	0

Spherical and Plane Waves (1 of 2)



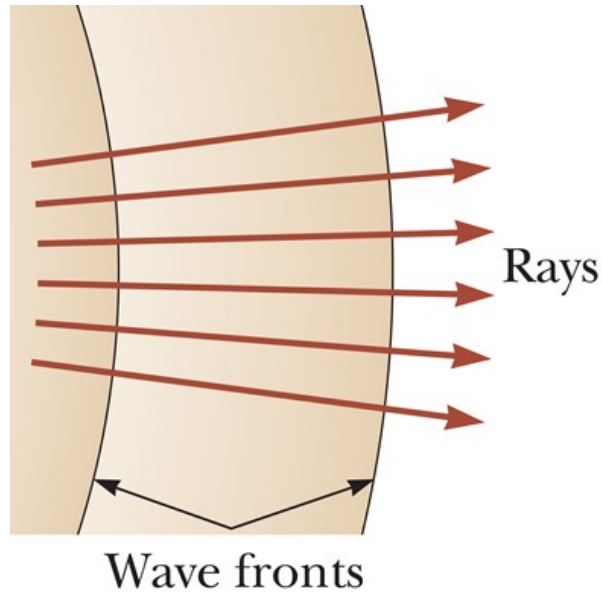
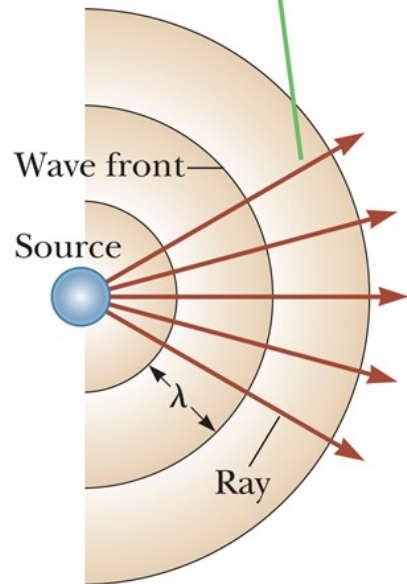
$$I = \frac{\text{average power}}{\text{area}} = \frac{P_{\text{av}}}{A} = \frac{P_{\text{av}}}{4\pi r^2}$$

$$I_1 = \frac{P_{\text{av}}}{4\pi r_1^2} \quad \text{and} \quad I_2 = \frac{P_{\text{av}}}{4\pi r_2^2}$$

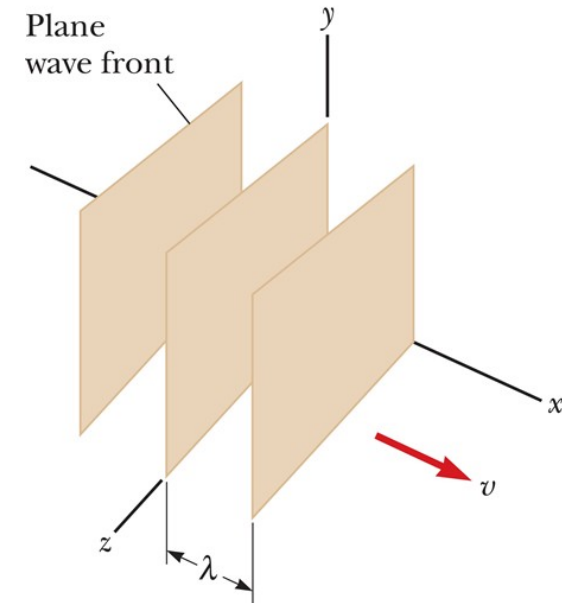
$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$$

Spherical and Plane Waves (2 of 2)

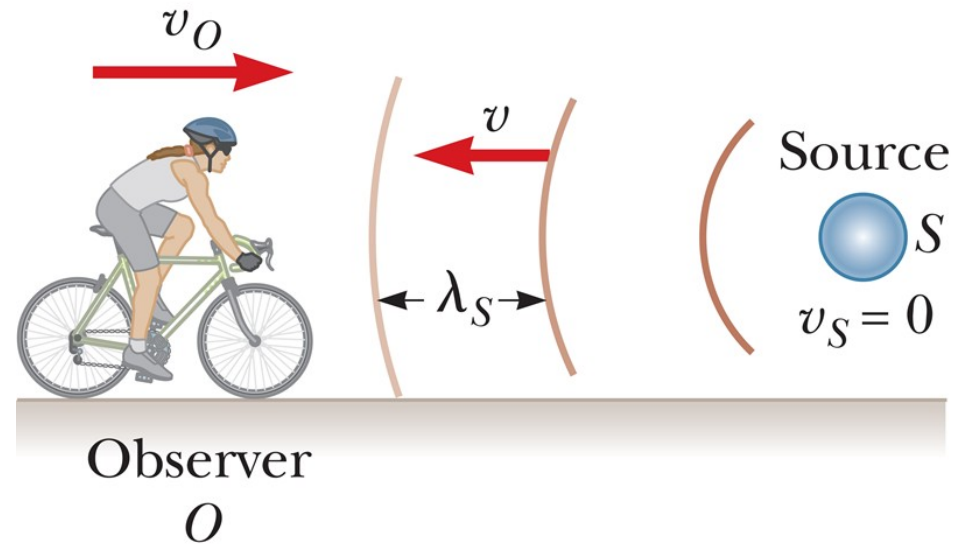
The rays are radial lines pointing outward from the source, perpendicular to the wave fronts.



The wave fronts are planes parallel to the yz -plane.



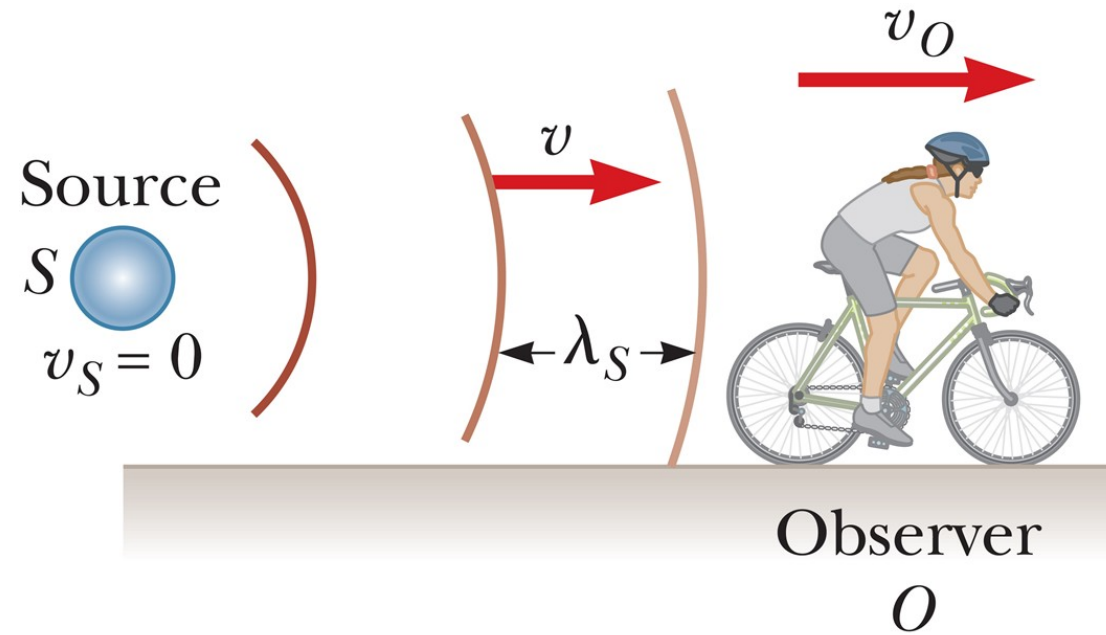
Case 1: The Observer Is Moving Relative to a Stationary Source (1 of 2)



$$\text{Additional wave fronts detected} = \frac{v_O t}{\lambda_S} \qquad f_O = f_S + \frac{v_O}{\lambda_S}$$

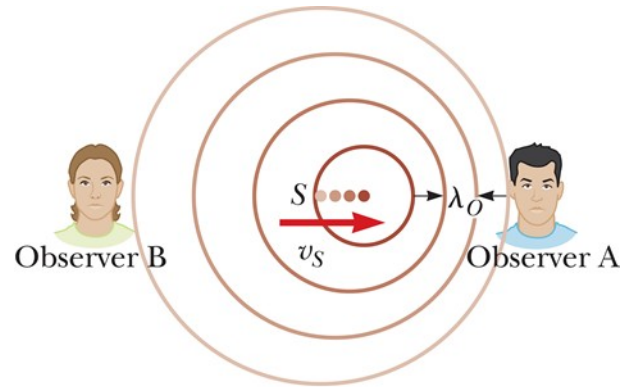
$$\lambda_S = \frac{v}{f_S} \rightarrow f_O = f_S \left(\frac{v + v_O}{v} \right) \quad (\text{observer moving } \textit{toward} \text{ source})$$

Case 1: The Observer Is Moving Relative to a Stationary Source (2 of 2)



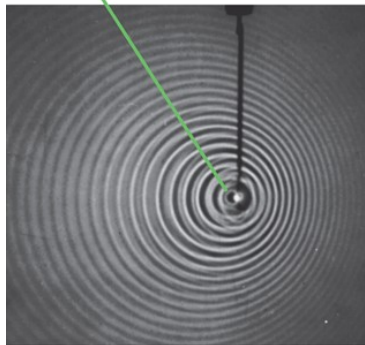
$$f_O = f_S \left(\frac{v - v_O}{v} \right) \quad (\text{observer moving away from source})$$

Case 2: The Source Is Moving Relative to a Stationary Observer



a

The source producing the water waves is moving to the right.



b

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$$\lambda_O = \lambda_S - \frac{v_S}{f_S} \quad \text{and} \quad f_O = \frac{v}{\lambda_O}$$

$$f_O = \frac{v}{\lambda_S - \frac{v_S}{f_S}} = \frac{vf_S}{\lambda_S f_S - v_S}$$

$$\lambda_S = v/f_S \quad \rightarrow \quad f_O = \frac{vf_S}{v - v_S}$$

$$f_O = f_S \left(\frac{v}{v - v_S} \right)$$

General Case

$$f_O = f_S \left(\frac{v + v_O}{v - v_S} \right)$$

Think – Pair – Share 2

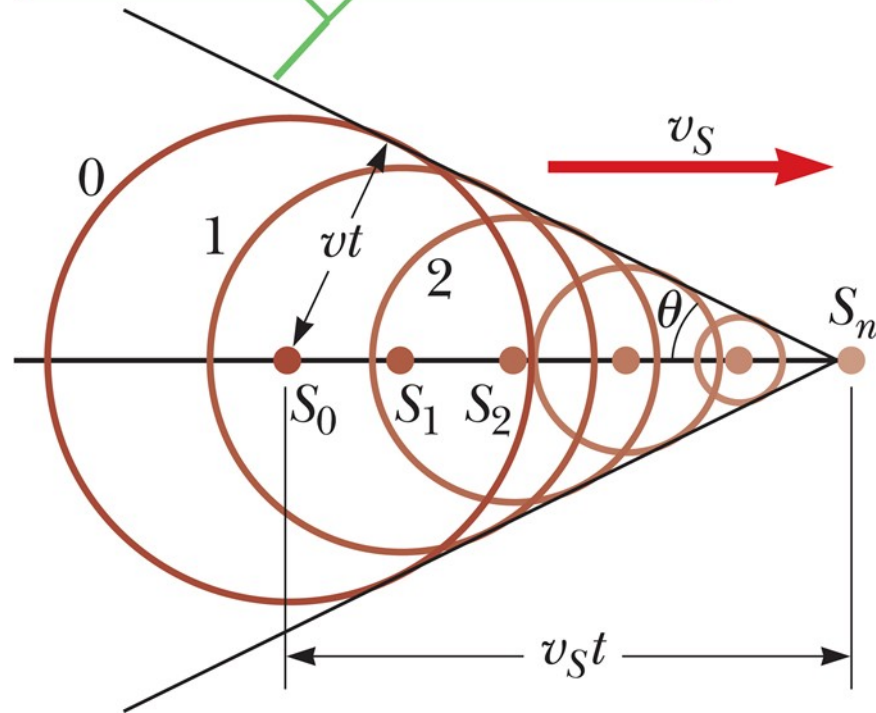
Suppose you're on a hot air balloon ride, carrying a buzzer that emits a sound of frequency f . If you accidentally drop the buzzer over the side while the balloon is rising at constant speed, what can you conclude about the sound you hear as the buzzer falls towards the ground?

1. The frequency and intensity increase.
2. The frequency decreases and the intensity increases.
3. The frequency decreases and the intensity decreases.
4. The frequency remains the same, but the intensity decreases.

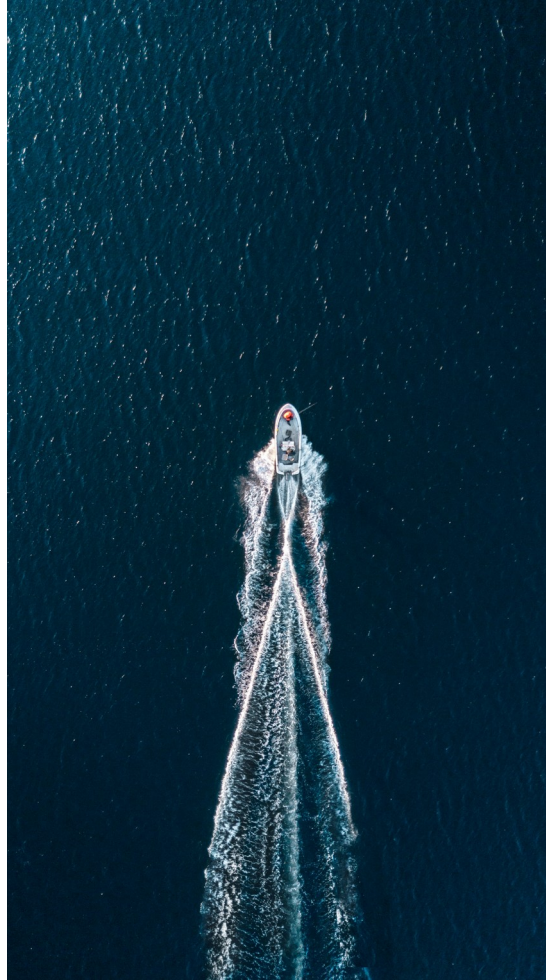
Shock Waves (1 of 3)

The envelope of the wave fronts forms a cone with half-angle of $\sin \theta = v/v_S$.

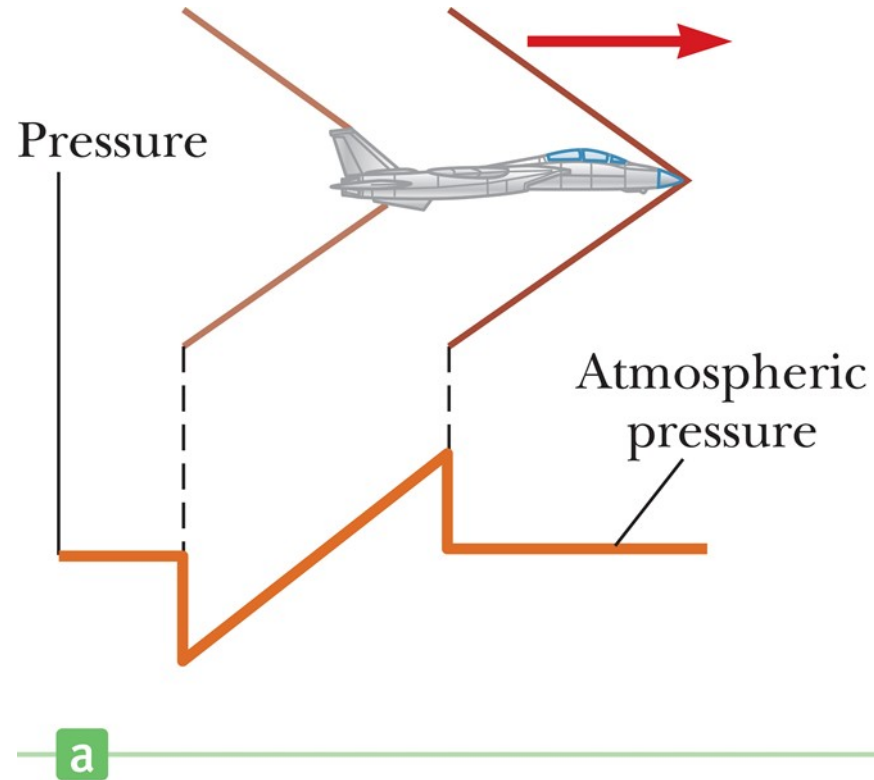
$$\sin \theta = \frac{v}{v_S}$$



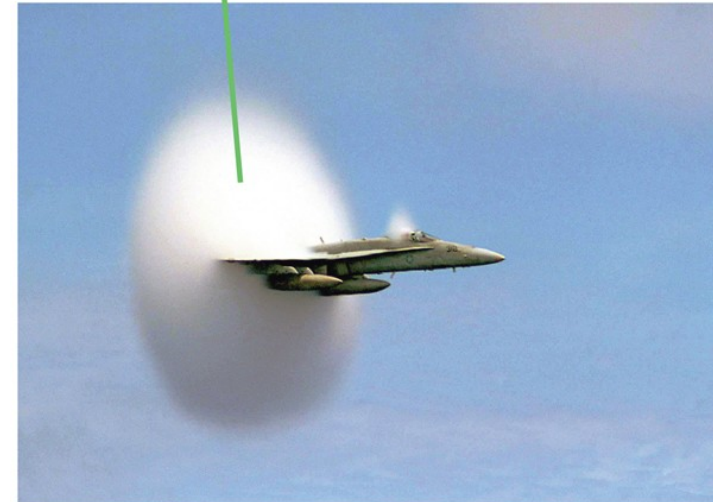
Shock Waves (2 of 3)



Shock Waves (3 of 3)



The large pressure variation in the shock wave condenses water vapor into droplets.



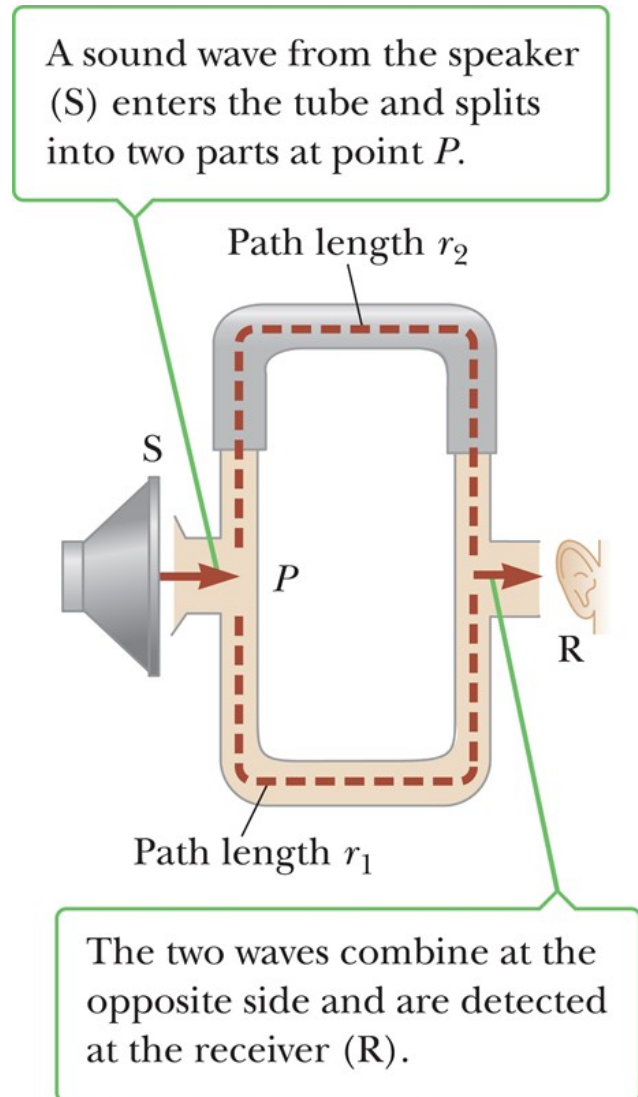
Keith Lawson/Bettmann/Corbis

Think – Pair – Share 3

As an airplane flying with constant velocity moves from a cold air mass into a warm air mass, the Mach number

1. increases.
2. decreases.
3. remains the same.

Interference of Sound Waves



constructive interference:

$$r_2 - r_1 = n\lambda \quad (n = 0, 1, 2, \dots)$$

destructive interference:

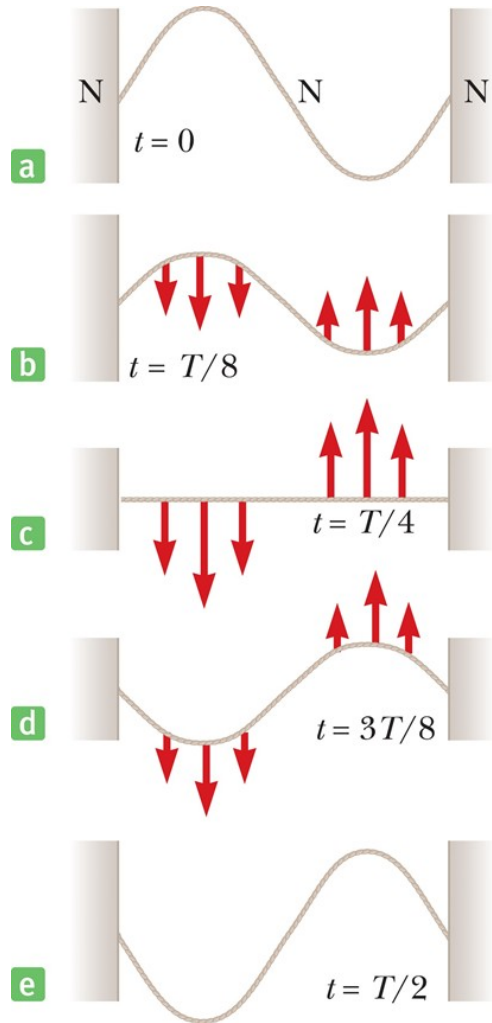
$$r_2 - r_1 = \left(n + \frac{1}{2}\right) \lambda \quad (n = 0, 1, 2, \dots)$$

Standing Waves (1 of 7)

Large-amplitude standing waves result when the blade vibrates at a natural frequency of the string.

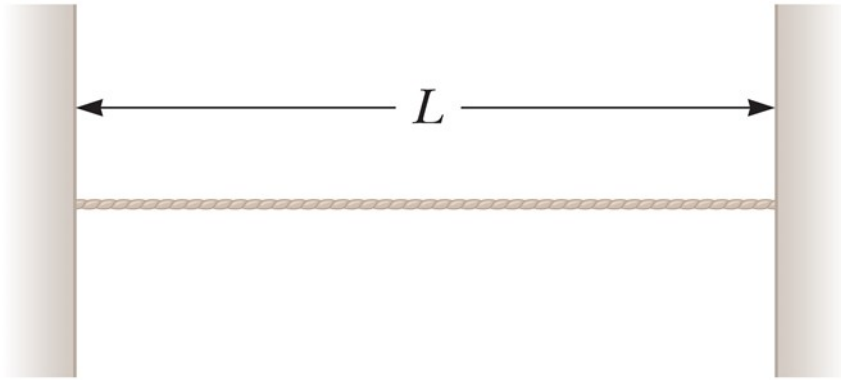


Standing Waves (2 of 7)



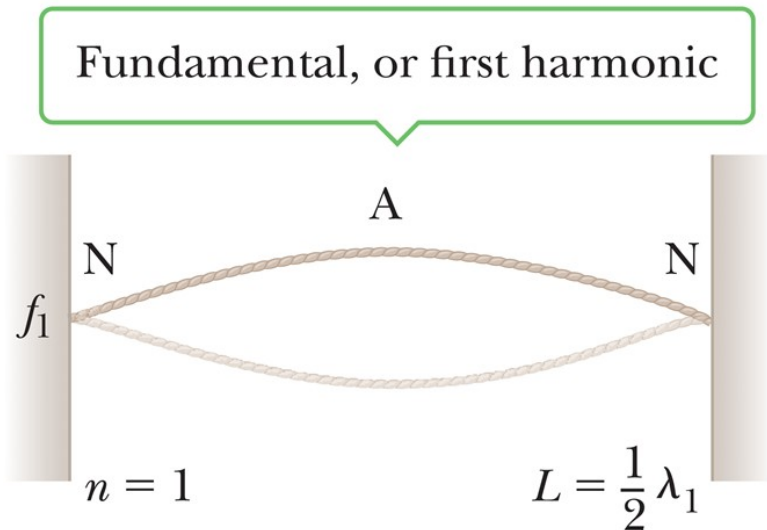
$$d_{\text{NN}} = \frac{1}{2}\lambda$$

Standing Waves (3 of 7)



$$L = 2 \left(\frac{\lambda_1}{4} \right) = \frac{\lambda_1}{2} \rightarrow \lambda_1 = 2L$$

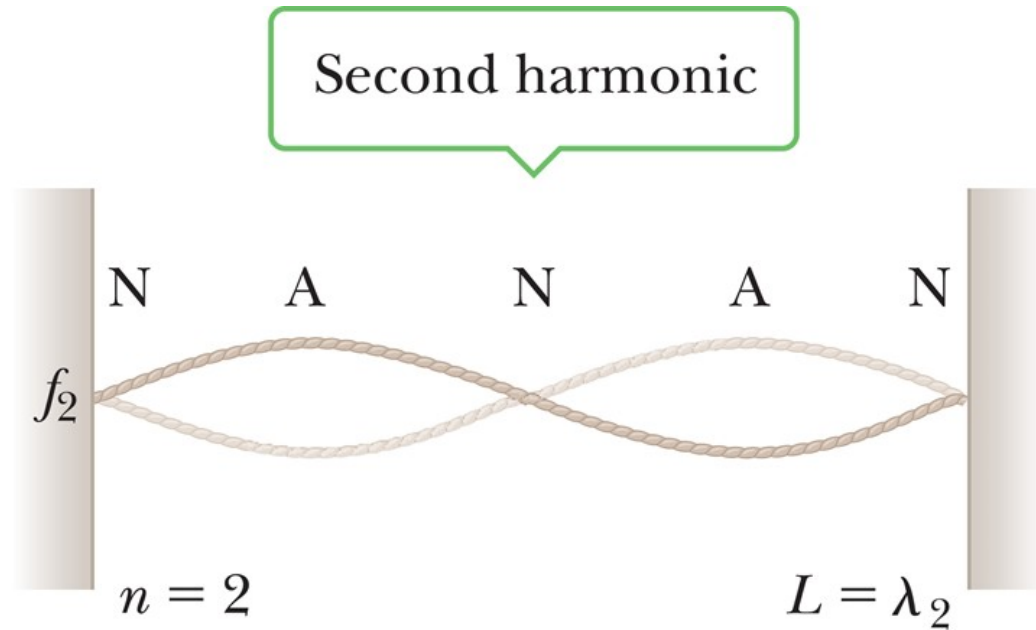
$$f_1 = \frac{v}{\lambda_1} = \frac{v}{2L}$$



$$v = \sqrt{\frac{F}{\mu}}$$

$$f_1 = \frac{1}{2L} \sqrt{\frac{F}{\mu}}$$

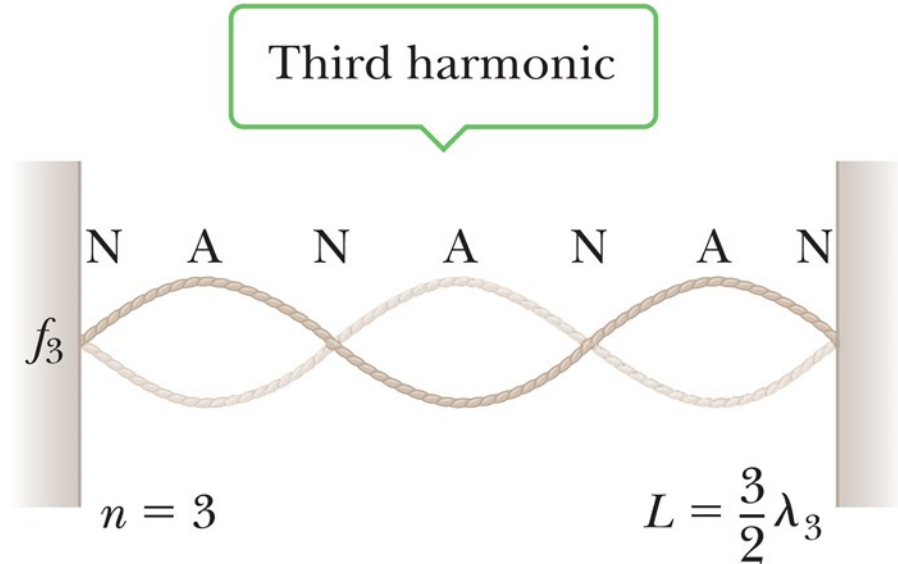
Standing Waves (4 of 7)



$$L = 4 \left(\frac{\lambda_2}{4} \right) = \lambda_2$$

$$f_2 = \frac{v}{\lambda_2} = \frac{v}{L} = 2 \left(\frac{v}{2L} \right) = 2f_1$$

Standing Waves (5 of 7)



$$L = 6 \left(\frac{\lambda_3}{4} \right) = 3 \left(\frac{\lambda_3}{2} \right)$$

$$f_3 = \frac{v}{\lambda_3} = \frac{3v}{2L} = 3f_1$$

$$f_n = nf_1 = \frac{n}{2L} \sqrt{\frac{F}{\mu}} \quad (n = 1, 2, 3, \dots)$$

Think – Pair – Share 4

Which of the following frequencies are higher harmonics of a string with fundamental frequency of 150 Hz?

a) 200 Hz

b) 300 Hz

c) 400 Hz

d) 500 Hz

e) 600 Hz

1. a

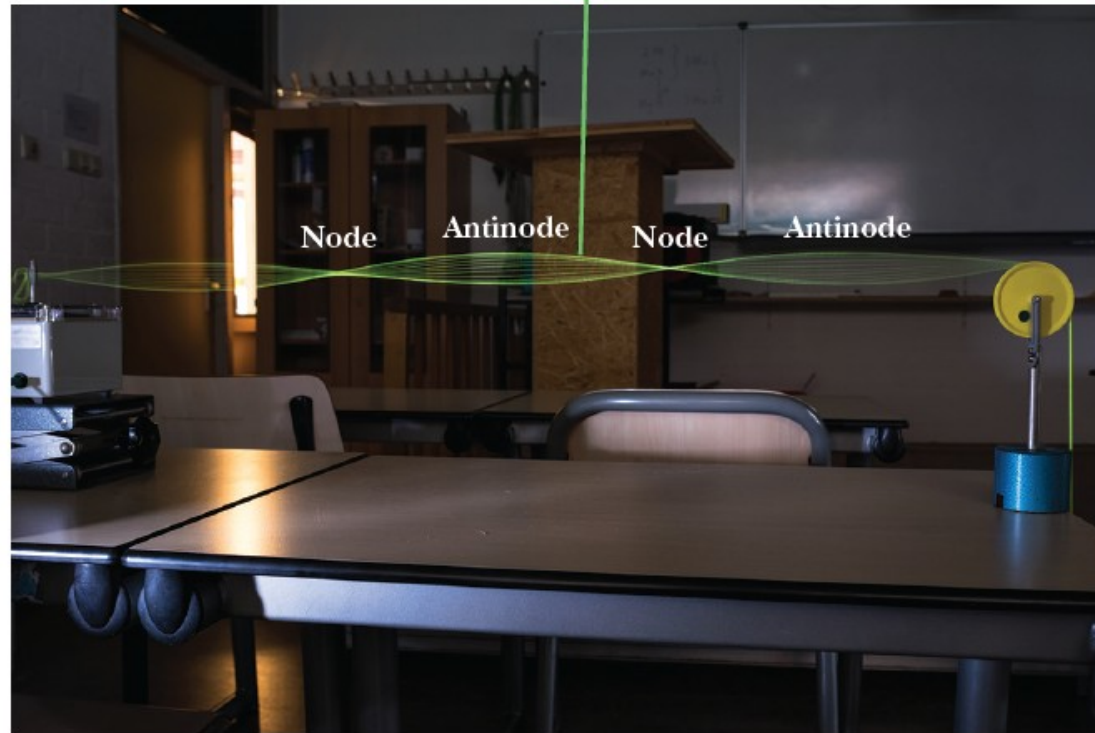
2. b and e

3. a, c, and e

4. d

Standing Waves (6 of 7)

The amplitude of the vertical oscillation of any element of the string depends on the horizontal position of the element.

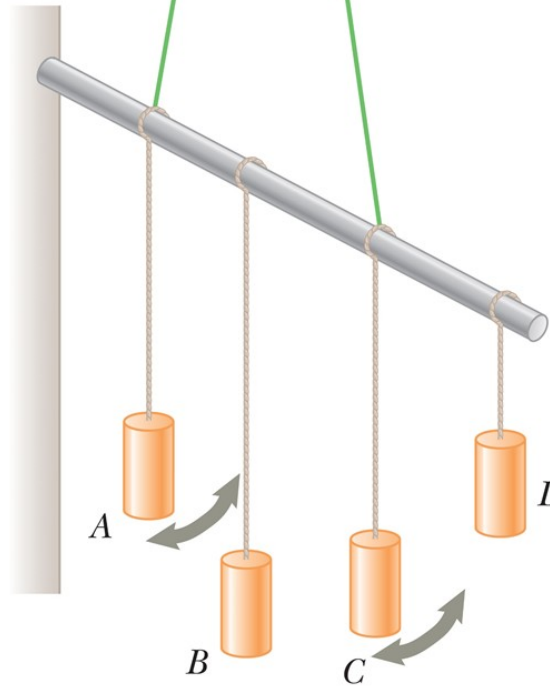


Standing Waves (7 of 7)



Forced Vibrations and Resonance

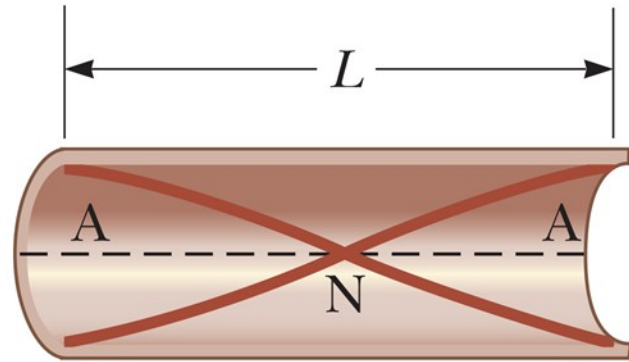
If pendulum *A* is set in oscillation, only pendulum *C*, with a length matching that of *A*, will eventually oscillate with a large amplitude, or resonate.



Standing Waves in Air Columns (1 of 3)

A pipe open at both ends.

First harmonic



$$\lambda_1 = 2L$$

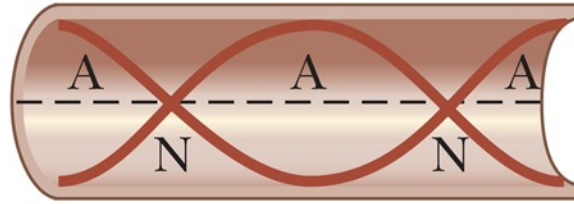
$$f_1 = \frac{v}{\lambda_1} = \frac{v}{2L}$$

$$L = 2 \left(\frac{\lambda_1}{4} \right) = \frac{\lambda_1}{2}$$

$$f = \frac{v}{\lambda_1} = \frac{v}{2L}$$

Standing Waves in Air Columns (2 of 3)

Second harmonic

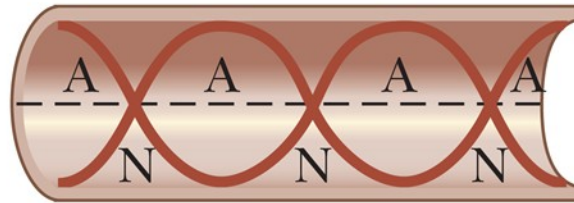


$$\lambda_2 = L$$

$$f_2 = \frac{v}{L} = 2f_1$$

$$f_n = n \frac{v}{2L} = nf_1 \quad (n = 1, 2, 3, \dots)$$

Third harmonic

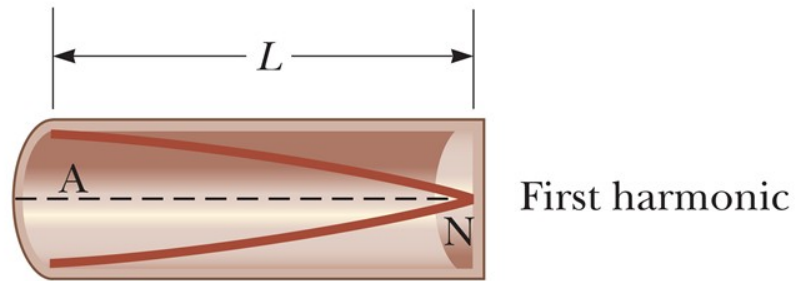


$$\lambda_3 = \frac{2}{3} L$$

$$f_3 = \frac{3v}{2L} = 3f_1$$

Standing Waves in Air Columns (3 of 3)

A pipe closed at one end.



$$\lambda_1 = 4L$$
$$f_1 = \frac{v}{\lambda_1} = \frac{v}{4L}$$



$$\lambda_3 = \frac{4}{3}L$$
$$f_3 = \frac{3v}{4L} = 3f_1$$

$$L = \frac{\lambda_1}{4} \rightarrow \lambda_1 = 4L$$

$$L = 3 \left(\frac{\lambda_3}{4} \right) \rightarrow \lambda_3 = \frac{4L}{3}$$

$$f_n = n \frac{v}{4L} = n f_1 \quad (n = 1, 3, 5, \dots)$$

Think – Pair – Share 5

A pipe open at both ends resonates at a fundamental frequency f_{open} . When one end is covered and the pipe is again made to resonate, the fundamental frequency is f_{closed} . Which of the following expressions describes how these two resonant frequencies compare?

1. $f_{\text{closed}} = f_{\text{open}}$

2. $f_{\text{closed}} = \frac{3}{2} f_{\text{open}}$

3. $f_{\text{closed}} = 2f_{\text{open}}$

4. $f_{\text{closed}} = \frac{1}{2} f_{\text{open}}$

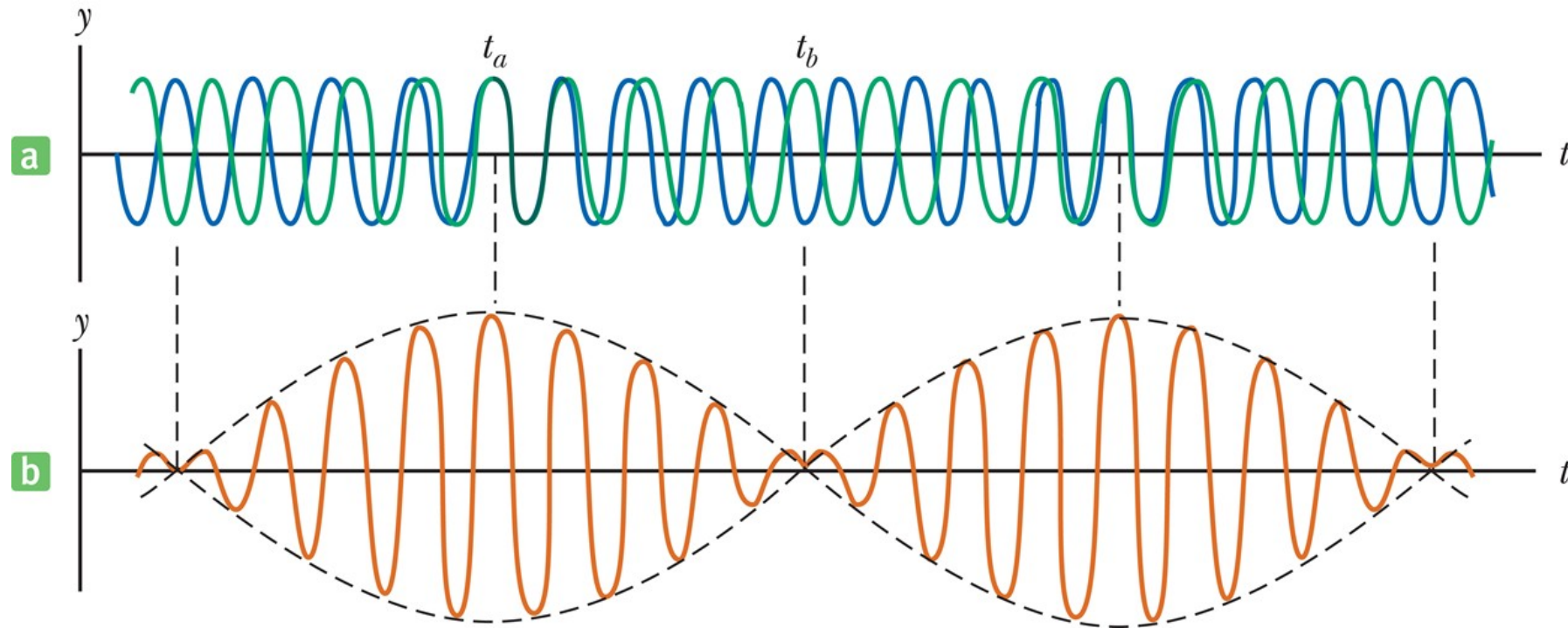
5. none of these

Think – Pair – Share 6

Balboa Park in San Diego has an outdoor organ. When the air temperature increases, the fundamental frequency of one of the organ pipes (the thermal expansion of the pipe is negligible)

1. increases.
2. decreases.
3. stays the same.
4. is impossible to determine.

Beats



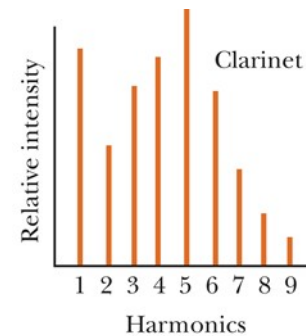
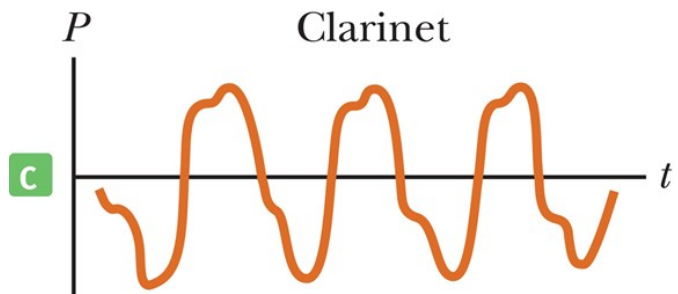
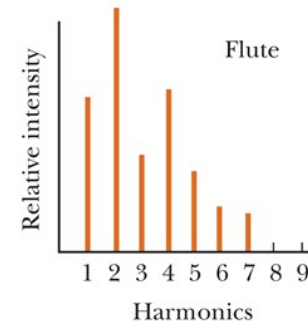
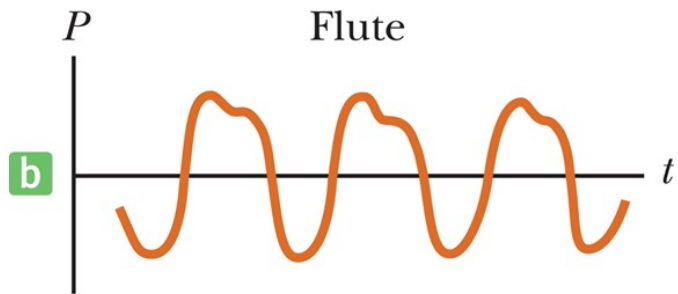
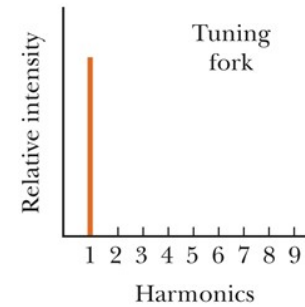
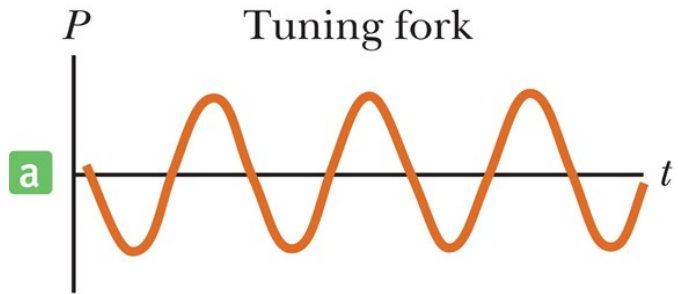
$$f_b = |f_2 - f_1|$$

Think – Pair – Share 7

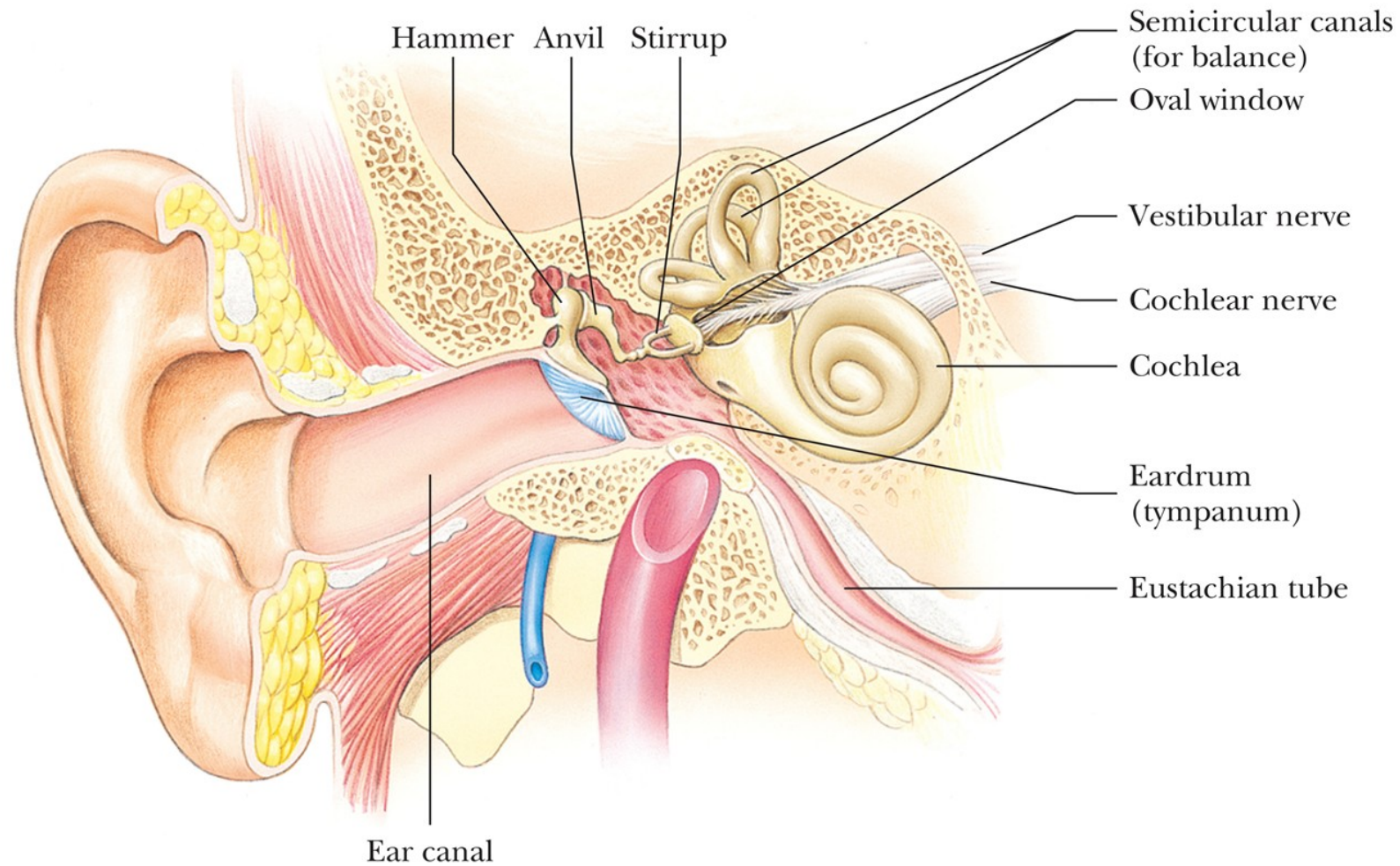
You are tuning a guitar by comparing the sound of the string with that of a standard tuning fork. You notice a beat frequency of 5 Hz when both sounds are present. You tighten the guitar string and the beat frequency rises steadily to 8 Hz. To tune the string exactly to the tuning fork, you should

1. continue to tighten the string.
2. loosen the string.
3. impossible to determine from the given information

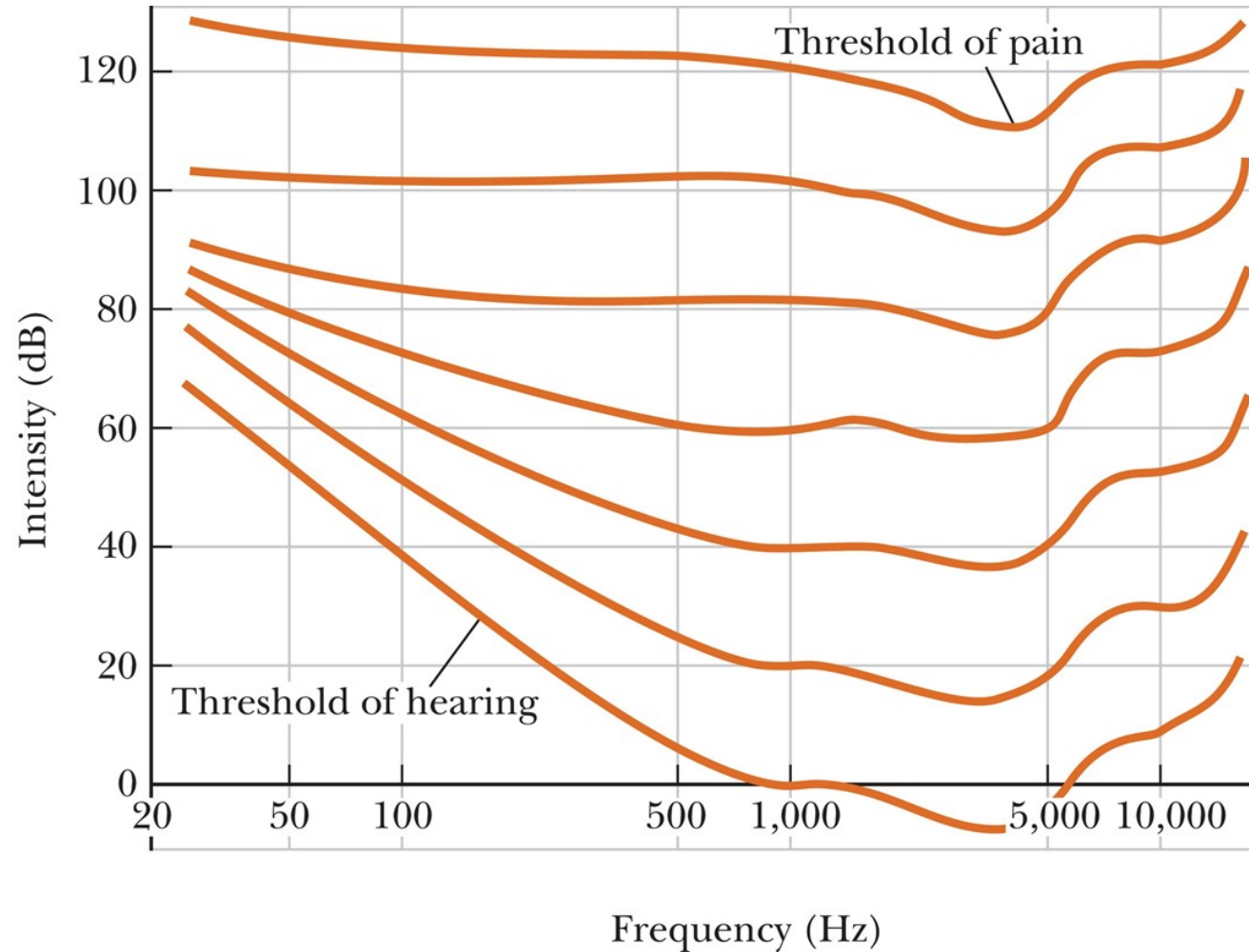
Quality of Sound



The Ear (1 of 3)



The Ear (2 of 3)



The Ear (3 of 3)

